Unit 2 · Session 1 · Tue, Oct 27, 2026

Musculoskeletal Pharmacology — Session 1

4 parts · every fact cited to its slide or textbook page. About 120 minutes of focused study.

Study plan

15 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

40 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

25 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

35 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. Tissue injury releases **prostaglandins, bradykinin, histamine, and cytokines** that sensitize nociceptors; **corticosteroids** block phospholipase A_2 upstream, while **NSAIDs** block COX. S6; S8; SG U2-R1; SG U2-R2

2. **Acetaminophen** treats pain and fever but is not a reliable anti-inflammatory; overdose depletes glutathione so **NAPQI** injures the liver, and **N-acetylcysteine** is the antidote. S10; S11; SG U2-R3

3. **COX-1** is the housekeeping enzyme (gastric protection, renal blood flow) and **COX-2** makes inflammatory prostaglandins; **celecoxib** lowers GI risk but raises cardiovascular risk, with similar renal risk. S15; S17; SG U2-R4; SG U2-R5

4. NSAID harms are **GI, renal, and cardiovascular**; flag anticoagulants, ACE inhibitors/ARBs plus diuretics (**triple whammy**), and corticosteroids — and prefer acetaminophen when red flags stack up. S19; S20; S21

5. **OA** is degenerative (treat pain and function); **RA** is autoimmune and needs **DMARDs**, which change the disease course; **methotrexate** is the anchor — **weekly**, with folate, never in pregnancy. S23; S25; S27; SG U2-R6; SG U2-R7

6. **Biologics** target the cascade — TNF (adalimumab, etanercept, infliximab), IL-6 (tocilizumab), B cells (rituximab), T-cell costimulation (abatacept) — and all DMARDs share **infection risk**: screen for TB and hepatitis, vaccinate first. S28; S30; SG U2-R8

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO2.1 Identify major pharmacologic classes used in the management of musculoskeletal conditions and describe their mechanisms of action.

Part 1 · Musculoskeletal Pain & the Inflammatory Cascade
Part 2 · Acetaminophen
Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat musculoskeletal conditions.

Part 2 · Acetaminophen
Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.4 Select appropriate pharmacologic therapy for common musculoskeletal conditions based on clinical presentation and patient-specific factors.

Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in musculoskeletal conditions.

Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for musculoskeletal.

Part 2 · Acetaminophen
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

SO1 Explain the pharmacologic basis of musculoskeletal pain and inflammation.

Part 1 · Musculoskeletal Pain & the Inflammatory Cascade

SO2 Compare acetaminophen and NSAIDs, including COX-1 vs. COX-2 inhibition.

Part 2 · Acetaminophen
Part 3 · NSAIDs & COX-2 Selective Inhibition

SO3 Identify major NSAID adverse effects and select therapy based on patient risk factors.

Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

SO4 Explain the mechanisms of conventional vs. biologic DMARDs, including representative generic agents in each class.

Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

SO5 Identify key contraindications, adverse effects, and drug interactions for analgesics and DMARDs.

Part 2 · Acetaminophen
Part 3 · NSAIDs & COX-2 Selective Inhibition
Part 4 · Osteoarthritis, Rheumatoid Arthritis & DMARDs

SO6 Preview: gout, skeletal muscle relaxants, and bone pharmacology continue in Session 2 (Thursday, October 29).

⚠ Preview only — taught in Session 2 (L04)

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ABOUT THIS LECTURE

- PA Pharmacology Course · Lecture 2 · Session 1 of 2 — Musculoskeletal Pharmacology (Tuesday, October 27): pain, acetaminophen, NSAIDs, and osteoarthritis/rheumatoid arthritis pharmacotherapy, including DMARDs. S1; S2 (notes)
- Session 2 (Thursday, October 29) picks up with gout, skeletal muscle relaxants, bone pharmacology, and the cumulative Lecture Recall Quiz. S2 (notes); S31
- Required text: Katzung's Basic and Clinical Pharmacology, 16th Edition (Todd W. Vanderah, 2025, McGraw-Hill Education; ISBN 1260463303 eBook; on AccessMedicine: accessmedicine.mhmedical.com/book.aspx?bookid=3382). Unit 2 — Musculoskeletal: Chapters 27, 36, and 42. S3; S3 (notes)
- Recommended text (additional support, not a substitute): Pharmacotherapy Principles and Practice (Chisholm-Burns, Schwinghammer, Malone, Kolesar, Lee, 2022, McGraw-Hill Education; Print ISBN 9781260460278, eText ISBN 9781260460285) — supplemental support alongside Katzung's Chapters 27, 36, and 42 for Unit 2. S3; S3 (notes)
- **Original figures in this deck:** Figures 1–3 (The Arachidonic Acid Pathway: Where Analgesics Act; Nonselective NSAIDs vs. COX-2 Selective Inhibition; Biologic DMARDs: Where Each Class Intercepts the Immune Cascade) were generated by the course director using the AI tool Claude to illustrate standard, non-copyrighted pharmacologic concepts — original diagrams, not reproduced from Katzung or any other copyrighted source. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S32

SLIDE DISCREPANCIES

- **Methotrexate mechanism:** the slides and study guide say dihydrofolate reductase inhibition; the textbook says that effect is relatively minor at RA doses and credits AICAR transformylase. Answer course exams with the slide/study-guide version. S27; SG U2-M; p. 688

1 PART 1 · TOPIC

Musculoskeletal Pain & the Inflammatory Cascade

SO1 IO2.1

Key definitions

- **Nociceptive pain:** direct activation of pain receptors by tissue injury. S5
- **Inflammatory pain:** pain driven by inflammatory mediators. S5
- **Neuropathic pain:** pain from nerve injury or dysfunction. S5
- **Acute pain:** recent onset, often tied to an identifiable injury or flare. S5
- **Chronic pain:** persists beyond expected healing time, often > 3 months. Treatment approach differs sharply between acute and chronic pain. S5

How it works

- **Tissue injury → release of inflammatory mediators → sensitization of nociceptors → pain.** S6
- Key mediators: **prostaglandins, bradykinin, histamine, and cytokines (TNF-α, IL-1, IL-6).** S6; SG U2-R1
- Arachidonic acid → COX enzyme → prostaglandins → pain and inflammation: the spine of the acetaminophen, NSAID, and corticosteroid sections. S7; S7 (notes)
- **NSAIDs and COX-2 inhibitors act at the COX step; glucocorticoids act further upstream; cytokine inhibitors and analgesics act through separate, parallel pathways.** S7
- **Corticosteroids block phospholipase A_2** (membrane phospholipids → arachidonic acid), so a broad range of mediators falls. NSAIDs block only COX, so they lower prostaglandins specifically. S8; SG U2-R2
- TB **The immune response:** chronic inflammation involves eicosanoids, leukotrienes, cytokines, chemokines, and many immune cells; RA, SLE, vasculitis, and gout all stem from this cascade. p. 675
- TB The lipoxygenase branch of arachidonic acid makes leukotrienes (chemotaxis, bronchoconstriction, vascular permeability) — a separate branch from the COX step NSAIDs block. p. 675; p. 678

Drugs named as examples

- Corticosteroids (block phospholipase A_2) vs. NSAIDs and COX-2 inhibitors (block COX): two distinct points of intervention. S8; S8 (notes)

CLINICAL

- Most musculoskeletal pain in this lecture is a mix of nociceptive and inflammatory; pure neuropathic pain belongs to a separate pharmacology unit. S5 (notes)
- The mediator list is the roadmap for the lecture: NSAIDs target prostaglandins, biologic DMARDs target specific cytokines. S6 (notes)
- TB **Therapeutic strategies:** goal 1 is symptom relief and function (NSAIDs and DMARDs); goal 2 is slowing tissue damage, for which DMARDs are absolutely necessary. Response is tracked with indices such as DAS28 in RA. p. 675

TRAP

- Corticosteroids act **upstream** at phospholipase A_2, not at COX — that is why their effect is broader than an NSAID's. S8; SG U2-R2
- The lecture's drugs target nociceptive and inflammatory pain; neuropathic pain is a different unit. S5 (notes)

Step by step

How injury becomes pain S6

Tissue injury → Inflammatory mediators released → Nociceptors sensitized → Pain

The arachidonic acid pathway S7; S8

Membrane phospholipids → Phospholipase A_2 (blocked by corticosteroids) → Arachidonic acid → COX-1 / COX-2 (blocked by NSAIDs) → Prostaglandins → Pain, inflammation, fever

Figure 1 · The arachidonic acid pathway: where analgesics act. Original illustrative diagram: membrane phospholipids → arachidonic acid → COX-1/COX-2 → prostaglandins → pain, inflammation, fever. Corticosteroids block phospholipase A_2 upstream; NSAIDs and COX-2 inhibitors block the COX step itself. S8; S8 (notes)

Where each drug class intervenes

Drug class	Step it blocks	What falls
Corticosteroids	**Phospholipase A_2** (upstream)	A broad range of mediators
NSAIDs, COX-2 inhibitors	**COX** enzyme	Prostaglandins specifically
Biologic DMARDs (cytokine inhibitors)	Specific cytokines or immune cells	TNF-α, IL-6 signaling; B or T cells

How to tell them apart. Sources: S7; S8; S6 (notes); SG U2-R2

WHY IT HAPPENS

- Why are corticosteroids broader than NSAIDs? They act at phospholipase A_2, before arachidonic acid is released, so every mediator made from it falls — NSAIDs only stop the COX branch. S8; SG U2-R2

MNEMONIC (MEMORY AID)

- Mediators that make pain: **P-B-H-C** — **P**rostaglandins, **B**radykinin, **H**istamine, **C**ytokines (TNF-α, IL-1, IL-6). S6; SG U2-R1

HIGH-YIELD

- Injury → mediators → nociceptor sensitization → pain. S6
- Musculoskeletal pain is driven by inflammatory mediators (prostaglandins, bradykinin, histamine, cytokines) — NSAIDs and corticosteroids intervene at different points along this cascade. S31; S6; SG U2-R1
- Corticosteroids = phospholipase A_2 (broad); NSAIDs = COX (prostaglandins). S8; SG U2-R2

2 PART 2 · DRUG CLASS

Acetaminophen

SO2 SO5 IO2.1 IO2.2 IO2.6

Class

- Effective analgesia, with a narrow margin in overdose: treats pain and fever well, but is **not an anti-inflammatory drug** the way NSAIDs are. S9; S10 (notes)

Prototype + other drugs named

- Acetaminophen (the lecture's single non-NSAID analgesic). S10

Mechanism of action

- **Analgesic effect:** relieves mild-to-moderate pain, with a central mechanism that is still incompletely understood. S10
- **Antipyretic effect:** reduces fever via central effects on the hypothalamic temperature-regulating center. S10
- **Anti-inflammatory effect:** minimal at the periphery — unlike NSAIDs, acetaminophen is not a reliable anti-inflammatory agent. S10
- **TB** Textbook adds: a weak peripheral COX-1/COX-2 inhibitor whose pain relief also involves opioid, cannabinoid, and serotonergic systems. p. 694

Indications

- Reasonable first-line for mild osteoarthritis pain. S24
- Generally safer first-line than an NSAID in a patient with NSAID red flags (kidney disease, anticoagulation, older age). S21
- **TB** Preferred over aspirin in aspirin allergy, hemophilia, peptic ulcer history, or aspirin-induced bronchospasm; alone it is inadequate for inflammatory conditions such as RA. p. 694

Adverse effects (boxed warnings first)

- **Overdose: glutathione stores are depleted, NAPQI accumulates, and hepatocyte injury follows.** S11
- **TB** Ingesting 15 g may be fatal (centrilobular hepatic necrosis); take care with liver disease, heavy alcohol use, or malnutrition. p. 694

Contraindications

Not covered in materials

Drug interactions

- **Hidden overdose:** many combination cold, flu, and sleep products also contain acetaminophen — **unintentional double-dosing is a leading cause of acetaminophen-related liver injury.** S12
- **TB** Combining acetaminophen with an NSAID improves pain control but can potentiate the NSAID's GI toxicity. p. 694

Pharmacokinetics

- Normal handling: acetaminophen → NAPQI (toxic metabolite) → glutathione conjugation → safe excretion. S11
- NAPQI = N-acetyl-p-benzoquinone imine — the "toxic metabolite" example from Lecture 1's drug-metabolism section. S11 (notes)
- **TB** Mostly conjugated to inactive sulfate and glucuronide; half-life 2–3 hours, relatively unaffected by renal function. p. 694

Monitoring

- **Always ask about ALL sources of acetaminophen.** S12
- Patients often don't realize a product contains acetaminophen unless they check the label — reinforce generic-name awareness. S12 (notes)
- **TB** Adults: generally no more than 4 g/day — and even 4 g is associated with more liver function test abnormalities. p. 694

CLINICAL PEARL / WHY

- **Antidote: N-acetylcysteine**, a glutathione precursor — it replenishes glutathione to detoxify NAPQI. S11; SG U2-R3
- Case: a patient on regular acetaminophen for back pain starts an OTC cold-and-flu combination. **What is the hidden danger in this combination?** Double-dosing acetaminophen. S12
- **TB** **Other analgesics — tramadol:** weak μ-opioid agonist plus serotonin/norepinephrine reuptake inhibition; no anti-inflammatory effect; up to 300 mg/day; risks serotonin syndrome and seizures. p. 694; p. 695

Step by step

Acetaminophen overdose S11; SG U2-R3

Acetaminophen → NAPQI (toxic metabolite) →

Glutathione conjugation → safe excretion → Overdose: glutathione depleted →

NAPQI accumulates → hepatocyte injury →

Antidote: N-acetylcysteine refills glutathione

Acetaminophen vs. NSAIDs

	Acetaminophen	NSAIDs
Pain	Yes (mild–moderate)	Yes
Fever	Yes (hypothalamus)	Yes
Inflammation	**Minimal — not reliable**	**Yes — powerful**
Main danger	Liver (NAPQI) in overdose or double-dosing	GI, renal, cardiovascular
Antidote	N-acetylcysteine	Not covered in materials

How to tell them apart. Sources: S10; S11; S13; S19

WHY IT HAPPENS

- Why does N-acetylcysteine work? Glutathione is what safely conjugates NAPQI. Overdose uses it up; N-acetylcysteine is a glutathione precursor, so it refills the supply. S11; SG U2-R3

MNEMONIC (MEMORY AID)

- Acetaminophen does **P**ain and **P**yrexia, not **P**uffiness (inflammation). S10

HIGH-YIELD

- Acetaminophen treats pain and fever but is not a reliable anti-inflammatory agent; its overdose toxicity is mediated by NAPQI accumulation. S31; S10; S11
- Overdose: glutathione depleted → NAPQI → hepatocyte injury. S11
- Antidote: N-acetylcysteine (glutathione precursor). S11; SG U2-R3
- Ask about ALL sources — combination cold/flu/sleep products hide acetaminophen. S12

3 PART 3 · DRUG CLASS

NSAIDs & COX-2 Selective Inhibition

SO2 SO3 SO5 IO2.1 IO2.2 IO2.3 IO2.4

Class

- Powerful anti-inflammatory effect, with real systemic risk. S13
- **Nonselective NSAIDs inhibit both COX-1 and COX-2.** S16
- **Celecoxib selectively inhibits COX-2 over COX-1.** S17

Prototype + other drugs named

- Ibuprofen — common first-line choice for mild-moderate pain. S16
- Naproxen — longer half-life, twice-daily dosing. S16
- Diclofenac — available in oral and topical formulations. S16
- Indomethacin — more potent, more GI/CNS adverse effects. S16
- **Ketorolac — short-term use only, significant GI/renal risk.** S16
- Celecoxib — the COX-2 selective agent. S17

Mechanism of action

- NSAID → COX inhibition → ↓ prostaglandins → ↓ pain, inflammation, fever. S14
- **COX-1, the "housekeeping" enzyme:** maintains gastric mucosal protection and normal renal blood flow; inhibiting it causes GI and renal adverse effects. S15; SG U2-R4
- **COX-2:** induced mainly at sites of inflammation and the primary source of inflammatory prostaglandins; selective inhibition targets inflammation with less GI toxicity. S15; SG U2-R4
- Simplified teaching model, not an absolute separation: COX-1 and COX-2 functions overlap, and COX-2 is also constitutively expressed in the kidney and elsewhere. S15 (notes)
- TB Aspirin irreversibly blocks platelet COX (the effect lasts the 8–10-day life of the platelet); other nonselective NSAIDs are reversible inhibitors. p. 675; p. 677

Indications

- Oral NSAIDs for osteoarthritis — effective, but carry the systemic risks below. S24
- Topical NSAIDs — effective local relief with much lower systemic exposure. S24
- TB NSAIDs ease symptoms but, unlike DMARDs, do not alter the course of any arthritic disorder. p. 676
- TB Used in RA, spondyloarthropathies, OA, sprains and strains, low back pain, and gout (tolmetin appears ineffective in gout). p. 676

- TB COX-2 inhibitors spare platelet thromboxane (made by COX-1) but block endothelial prostacyclin — so they are not cardioprotective. p. 677

Adverse effects (boxed warnings first)

- **Gastrointestinal:** gastritis, peptic ulcer, GI bleeding. S19
- **Renal:** decreased renal perfusion, fluid retention, acute kidney injury. S19
- **Cardiovascular:** hypertension, increased risk of cardiovascular events. S19
- **Celecoxib:** lower risk of ulcer/bleeding, but **COX-2 selectivity is associated with increased cardiovascular event risk — a key trade-off.** S17; SG U2-R5
- Celecoxib's renal risk is similar to nonselective NSAIDs — COX-2 selectivity does not protect the kidney. S17; SG U2-R5
- TB Beyond GI, renal, and CV: headache, tinnitus, dizziness, rare aseptic meningitis, liver test abnormalities, asthma, rashes, hyperkalemia. p. 676

Contraindications

- Use caution or avoid in: active GI bleeding or ulcer history; chronic kidney disease; heart failure or cardiovascular disease. S20
- Also caution or avoid with anticoagulant therapy and in late pregnancy. S20
- TB Ibuprofen is relatively contraindicated with nasal polyps, angioedema, or bronchospastic reactivity to aspirin. p. 680

Drug interactions

- NSAIDs + anticoagulants → ↑ bleeding risk. S20
- NSAIDs + ACE inhibitors/ARBs → ↑ AKI risk; add a diuretic and it is the **"triple whammy,"** a very high-yield AKI pattern. S20; S20 (notes)
- NSAIDs + diuretics → reduced diuretic effect, ↑ renal risk. S20
- NSAIDs + corticosteroids → ↑ GI ulcer risk. S20
- TB Ibuprofen taken with aspirin blocks aspirin's irreversible platelet inhibition, limiting its cardioprotective effect. p. 680
- TB Celecoxib (a CYP2C9 substrate) occasionally interacts with warfarin and raises lithium and fluconazole levels. p. 679

Pharmacokinetics

- Naproxen's longer half-life allows twice-daily dosing; diclofenac comes as oral and topical forms. S16
- TB **Chemistry & pharmacokinetics:** most NSAIDs are weak acids, well absorbed, ~98% albumin-bound, metabolized by CYP3A or CYP2C, and renally excreted. p. 675

Monitoring

- These three systems (GI, renal, CV) are the organizing framework for every NSAID contraindication and interaction discussion (Table 1). S19 (notes)
- TB Highest GI-bleeding risk: celecoxib, or a nonselective NSAID plus omeprazole or misoprostol. p. 681

CLINICAL PEARL / WHY

- Case: a 72-year-old with OA, CKD, hypertension, and an anticoagulant. **Is an NSAID a good first-line choice for this patient's pain? Why or why not?** S21
- **Three NSAID red flags at once** — renal disease, anticoagulation, and age-related bleeding risk — make acetaminophen or a non-NSAID strategy generally safer first-line. S21
- Celecoxib is a classic "trade one risk for another" point: a better GI profile, but not a free lunch cardiovascularly or renally. S17 (notes)
- There is no risk-free NSAID choice, only a different risk profile. S18 (notes)
- TB **Choice of NSAID:** all are about equally effective, so choose by toxicity, cost, and patient factors; in renal insufficiency nonacetylated salicylates may be best. p. 681

Step by step

How NSAIDs work S14

NSAID → COX inhibition → ↓ Prostaglandins →

↓ Pain, inflammation, fever

Is an NSAID the right first choice? S19; S20; S21; S24

Check GI: ulcer or bleeding history? → Check kidney: CKD, ACEi/ARB, diuretic?

→ Check heart: HF, CV disease, hypertension? →

Check bleeding: anticoagulant, older age? →

Red flags → acetaminophen or topical/non-NSAID strategy

Figure 2 · Nonselective NSAIDs vs. COX-2 selective inhibition. Original illustrative comparison (illustrative relative-risk values, not from Katzung): nonselective NSAIDs (e.g., ibuprofen, naproxen) carry higher GI ulcer/bleeding risk (75) than cardiovascular risk (35); celecoxib trades lower GI risk (30) for higher cardiovascular risk (65). S18; S18 (notes)

COX-1 vs. COX-2

	COX-1	COX-2
Role	**"Housekeeping"** enzyme	Induced mainly at sites of inflammation
Protects	Gastric mucosa; normal renal blood flow	Not covered in materials
Makes	Not covered in materials	Primary source of **inflammatory prostaglandins**
Blocking it causes	GI and renal adverse effects	Less inflammation (selective blockade = less GI toxicity)

How to tell them apart. Sources: S15; SG U2-R4

Nonselective NSAIDs vs. celecoxib

	Nonselective (ibuprofen, naproxen…)	Celecoxib (COX-2 selective)
GI ulcer / bleeding	**Higher** (Figure 2: 75)	**Lower** (Figure 2: 30)
Cardiovascular events	Lower (Figure 2: 35)	**Higher** (Figure 2: 65)

	Nonselective (ibuprofen, naproxen…)	Celecoxib (COX-2 selective)
Renal risk	Present	**Similar** — selectivity does not protect the kidney

How to tell them apart. Sources: S17; S18; SG U2-R5

Other NSAIDs in the reading (textbook)

Drug	Half-life (h)	What sets it apart
Aspirin	0.25	Now mainly antiplatelet (81–325 mg daily); avoid in hemophilia
Nonacetylated salicylates	Not covered in materials	Anti-inflammatory without platelet inhibition: an option in asthma, bleeding tendency, renal dysfunction
Ibuprofen	2	≤ 1600 mg/day analgesic only; 2400 mg/day anti-inflammatory
Naproxen	12–17	More GI bleeding than OTC ibuprofen, better CV profile; avoid in hepatic impairment
Diclofenac	1.9–2.3	More aminotransferase rises; 1% topical gel OTC
Indomethacin	4–5	Headache in 11.7%; can worsen depression, epilepsy, parkinsonism
Ketorolac	5–6	Analgesic use; start IV/IM/intranasal; ≤ 120 mg/day; ≤ 5 days
Celecoxib	11.2	10–20× COX-2 selective; 100–200 mg bid; no platelet effect at usual doses
Meloxicam	20	Relatively COX-2 selective at 7.5 mg/day; fewer GI effects
Etodolac	6.4	Relatively COX-2 selective; linked to acute pancreatitis
Nabumetone	22.5–29.8	Only nonacid NSAID (prodrug); less ulceration
Piroxicam	50	Above 20 mg/day sharply raises ulcer and bleeding risk
Sulindac	16.4	Less effect on renal prostaglandins; more liver test abnormalities
Oxaprozin · diflunisal · flurbiprofen · tolmetin	54.9 · 8–12 · 3.8 · 1–2	Oxaprozin once daily; diflunisal rarely used; flurbiprofen rare tremor/rigidity; tolmetin ineffective in gout

How to tell them apart. Sources: p. 677; p. 679; p. 680; p. 681; p. 694

WHY IT HAPPENS

- Why do NSAIDs hurt the stomach and kidney? Blocking COX-1, the housekeeping enzyme, removes the prostaglandins that protect gastric mucosa and keep renal blood flow normal. S15

- Why the “triple whammy”? An ACE inhibitor/ARB, a diuretic, and an NSAID each push kidney perfusion the wrong way at once — together they cause acute kidney injury. S20; S20 (notes)

MNEMONIC (MEMORY AID)

- NSAID harms = **GRC**: **G**ut, **R**enal, **C**ardiovascular (Table 1). S19

HIGH-YIELD

- COX-1 = housekeeping (gastric protection, renal blood flow); COX-2 = inflammation. S15; SG U2-R4
- NSAIDs are effective but carry real GI, renal, and cardiovascular risk; celecoxib trades GI risk for cardiovascular risk (renal risk similar). S31; S17; SG U2-R5
- NSAID harms: GI, renal, cardiovascular. S19
- Triple whammy: NSAID + ACEi/ARB + diuretic → AKI. S20 (notes)
- Ketorolac: short-term only; significant GI/renal risk. S16

4 PART 4 · DRUG CLASS

Osteoarthritis, Rheumatoid Arthritis & DMARDs

IO2.1 IO2.2 IO2.3 IO2.4 IO2.5 IO2.6 SO3 SO4 SO5

Class

- **Osteoarthritis (OA):** primarily degenerative joint disease — cartilage breakdown with mechanical wear. S23
- **Rheumatoid arthritis (RA):** autoimmune, systemic inflammatory disease — immune-mediated joint destruction. S23
- OA treatment goal: pain control and function. S23
- RA treatment goal: suppress disease activity, not just pain. S23
- The hinge: OA is managed mainly with analgesics/NSAIDs, while RA requires disease-modifying therapy. S23 (notes)
- **DMARD (Disease-Modifying Antirheumatic Drug):** a drug intended to alter the course of RA, not just mask its symptoms. S25; SG U2-R6
- **Goal of therapy:** slow or halt joint destruction and systemic inflammation, rather than providing analgesia alone. S25
- An NSAID can make a patient feel better without changing the disease trajectory; a DMARD is meant to change the trajectory itself. S25 (notes); SG U2-R6
- Targeted therapy: each biologic class corresponds to a different cytokine or cell type in the inflammatory cascade — the mediator list from the start of the lecture. S28 (notes)
- **TB** DMARD effects take 2 weeks to 6 months to appear. Types: conventional synthetic (cs), biologic (b), and targeted synthetic (ts, e.g., tofacitinib). p. 681
- **TB** Biologics are large molecules, usually proteins made by recombinant DNA technology; JAK inhibitors are oral small-

Prototype + other drugs named

- **Methotrexate — the anchor drug for most RA regimens.** S26
- Hydroxychloroquine — milder disease; relatively favorable safety profile. S26
- Sulfasalazine — often used in combination regimens. S26
- Leflunomide — alternative when methotrexate is not tolerated. S26
- Adalimumab, etanercept, infliximab — **TNF inhibitors**. S28; SG U2-R8
- Tocilizumab — **IL-6 inhibitor**. S28; SG U2-R8
- Rituximab — **B-cell targeted therapy**. S28; SG U2-R8
- Abatacept — **T-cell costimulation modulator**. S28; SG U2-R8

molecule targeted synthetic DMARDs. p. 681; p. 686

Mechanism of action

- **Mechanism: inhibits dihydrofolate reductase, reducing immune cell proliferation at low RA doses.** (methotrexate) S27; SG U2-M
- TNF inhibitors block tumor necrosis factor-alpha signaling. S28
- IL-6 inhibitors block interleukin-6 receptor signaling. S28
- B-cell targeted therapy depletes or modulates B lymphocytes. S28
- T-cell costimulation modulators block the second signal needed for T-cell activation. S28; SG U2-M
- Figure 3: abatacept acts at T-cell activation, TNF inhibitors and tocilizumab at cytokine release, rituximab at B cells — all upstream of synovial inflammation and joint destruction. S29

Indications

- **OA step 1:** acetaminophen — reasonable first-line for mild OA pain. S24
- **OA step 2:** oral NSAIDs (effective, systemic risks) or topical NSAIDs (local relief, much lower systemic exposure). S24
- **Intra-articular corticosteroid injections:** short-term relief for flares in a single joint. S24
- Other approaches: activity modification, physical therapy, weight management. S24
- Methotrexate anchors most RA regimens; leflunomide is the alternative when it is not tolerated. S26
- Rheumatoid arthritis: they modify the disease course rather than only masking symptoms. S31; SG U2-R6
- TB **Glucocorticoid drugs in rheumatic diseases:** used chronically in 30–70% of RA patients; relief is prompt and dramatic, and they slow new bone erosions. p. 693
- TB Long-term prednisone in RA: no more than 7.5 mg daily, tapered when possible — even 2.8 mg daily for over 5 years is linked to severe infections. p. 693
- TB Intra-articular corticosteroids often relieve painful symptoms for up to several months. p. 693
- TB Long-term glucocorticoid harms: osteoporosis, hyperglycemia, Cushing syndrome, infections, cataract, glaucoma, gastric ulcer, depression. p. 694
- TB Methotrexate starts at 7.5 mg weekly and rises to 15–25 mg weekly; it slows new erosions in RA. p. 688
- TB **Combination therapy with DMARDs** is now the rule when monotherapy falls short; triple therapy (methotrexate, sulfasalazine, hydroxychloroquine) rivals etanercept plus methotrexate. p. 693

Adverse effects (boxed warnings first)

- TB **Boxed warning — all TNF inhibitors:** possible malignancy, especially lymphoma. p. 692
- TB **Boxed warning — all JAK inhibitors:** venous thromboembolism (DVT, PE), major adverse cardiovascular events, and cancer. p. 686
- Methotrexate key adverse effects: **hepatotoxicity, bone marrow suppression, pulmonary toxicity.** S27; SG U2-R7
- **The major safety theme across all DMARDs: immunosuppression → increased infection risk.** This holds for conventional and biologic DMARDs alike. S30; SG U2-R8
- TB Methotrexate's most common toxicities are nausea and stomatitis; dose-related liver enzyme elevations are frequent. p. 689
- TB TNF inhibitors: TB reactivation, fungal infections, hepatitis B reactivation, new or worse heart failure, demyelinating disease, lupus-like syndrome. p. 690; p. 692

Contraindications

- **Methotrexate is contraindicated in pregnancy (teratogenic).** S27
- TB Don't stack biologics: abatacept with a TNF inhibitor probably raises serious infections; upadacitinib with biologic DMARDs is not recommended. p. 682; p. 687

Drug interactions

- TB Hydroxychloroquine can reduce methotrexate clearance. p. 688
- TB Tocilizumab stimulates CYP450 — doses of cyclosporine or warfarin may need adjusting. p. 692
- TB Methotrexate given with TNF inhibitors reduces anti-drug antibodies (infliximab, golimumab, certolizumab). p. 691; p. 692

Pharmacokinetics

- **Methotrexate is dosed weekly (not daily) — a critical patient-safety point, since daily dosing can be fatal.** S27; SG U2-R7

Monitoring

- **Folate supplementation** is routinely co-prescribed with methotrexate to reduce toxicity. S27; SG U2-R7

- **Baseline screening:** e.g., tuberculosis and hepatitis B/C screening before starting biologics. S30
- **Vaccination:** update vaccines, especially live vaccines, before starting therapy. S30
- **Ongoing monitoring:** regular labs to catch cytopenias, hepatotoxicity, or infection early. S30
- **Infection precautions:** counsel patients to report fevers or infection symptoms promptly. S30
- **TB** Folic acid or leucovorin reduces stomatitis, GI, and liver test problems but may cut methotrexate efficacy by 10–18%. p. 689
- **TB** Hydroxychloroquine: baseline eye exam, then annual screening after 5 years (retinopathy). p. 684
- **TB** Avoid live vaccines for up to 3 months after stopping abatacept; stop upadacitinib before a live attenuated vaccine. p. 682; p. 687

CLINICAL PEARL / WHY

- A 72-year-old with OA, CKD, hypertension, and an anticoagulant: acetaminophen or a non-NSAID strategy first. S21
- OA and RA both affect the joint but are two very different diseases — don't give RA patients analgesia alone. S22; S23 (notes)
- Methotrexate's weekly dosing is one of the highest-yield patient-safety points in all of rheumatology pharmacology. S27 (notes)
- Match each generic name to its class: adalimumab, etanercept, infliximab (TNF); tocilizumab (IL-6); rituximab (B cell); abatacept (T-cell costimulation). S28; S29; S31
- **TB** Topical diclofenac 1% gel is sold over the counter for OA and muscular pain. p. 679
- **TB** **Biosimilars:** near-copies of legacy biologics with the same efficacy, toxicity, and immunogenicity at potentially lower cost. p. 693
- **TB** If methotrexate alone falls short: add csDMARDs or a biologic — usually a TNF inhibitor first, then non-TNF biologics or JAK inhibitors. p. 701

Step by step

Methotrexate safety S27; SG U2-R7

Weekly dose (never daily) → Folate co-prescribed →

Watch liver, marrow, lung → Avoid in pregnancy (teratogenic)

Starting a biologic S30

Screen: TB, hepatitis B/C → Update vaccines (especially live) →

Start therapy → Regular labs: cytopenias, liver, infection →

Report fever or infection promptly

Figure 3 · Biologic DMARDs: where each class intercepts the immune cascade. Original illustrative diagram (not from Katzung): abatacept blocks T-cell costimulation; TNF inhibitors (adalimumab, etanercept, infliximab) and tocilizumab block cytokine signaling; rituximab depletes B cells — each upstream of synovial inflammation and joint destruction. S29

OA vs. RA at a glance

	Osteoarthritis	Rheumatoid arthritis
Disease type	Degenerative	**Autoimmune, systemic inflammatory**
What damages the joint	Cartilage breakdown, mechanical wear	Immune-mediated joint destruction
Treatment goal	Pain control and function	**Suppress disease activity**, not just pain
Core drugs	Acetaminophen, NSAIDs (oral/topical), intra-articular steroids	**DMARDs** (methotrexate first)

How to tell them apart. Sources: S23; S23 (notes); S24; S26

The four biologic classes in the lecture

Class	Target	Example
TNF inhibitors	Block TNF-α signaling	**Adalimumab, etanercept, infliximab**
IL-6 inhibitors	Block IL-6 receptor signaling	**Tocilizumab**
B-cell targeted therapy	Deplete or modulate B lymphocytes	**Rituximab**
T-cell costimulation modulator	Block the second signal for T-cell activation	**Abatacept**

How to tell them apart. Sources: S28; S29; SG U2-R8

Conventional synthetic DMARDs (textbook)

Drug	How it works	Main uses	Watch for
Methotrexate	DHFR (slides); AICAR transformylase (textbook)	RA anchor; psoriasis, PsA, JIA	Nausea, stomatitis, liver, marrow, lung; teratogenic
Hydroxychloroquine, chloroquine	Suppress T-cell responses and chemotaxis; dampen type 1 IFN	Mild RA; SLE (lowers mortality); Sjögren	Retinopathy (eye exams); cardiomyopathy; 3–6 months to respond
Sulfasalazine	Split into sulfapyridine + 5-aminosalicylic acid	RA (slows radiologic progression); ulcerative colitis	~30% stop it; rash, toxic epidermal necrolysis, neutropenia; reversible male infertility

Drug	How it works	Main uses	Watch for
Leflunomide	Dihydroorotate dehydrogenase inhibitor	RA (equals methotrexate)	Diarrhea (25%), liver tests, hypertension; half-life 19 days (cholestyramine shortens)
Azathioprine	Becomes 6-thioguanine; suppresses B and T cells	RA, SLE, transplant	Marrow suppression (TPMT, NUDT15), GI upset, cancer, infection
Mycophenolate mofetil	IMPDH inhibitor	SLE (renal and nonrenal), scleroderma lung; not proven in RA	GI, liver, cytopenias, infection; contraindicated in pregnancy
Cyclosporine, tacrolimus, voclosporin	Calcineurin inhibitors (block T-cell cytokines)	Severe RA; lupus nephritis	Nephro-, neuro-, cardiotoxicity, hypertension; CYP3A4 interactions (grapefruit)
Cyclophosphamide	Cross-links DNA	Lupus nephritis, CNS lupus, vasculitis	Cytopenias, infection, sterility, bladder and skin cancer
Apremilast	PDE4 inhibitor (↑ cAMP)	Plaque psoriasis, PsA, Behçet oral ulcers	Diarrhea, nausea, weight loss, mood changes; CYP3A4
Mesalamine	Topical colonic 5-aminosalicylate	Ulcerative colitis	Diarrhea, headache; renal impairment (monitor)

How to tell them apart. Sources: S26; S27; p. 682; p. 683; p. 684; p. 687; p. 688; p. 689; p. 690

Biologic and targeted DMARDs in the reading (textbook)

Drug(s)	Target	Main uses	Watch for
Adalimumab, etanercept, infliximab, certolizumab, golimumab	TNF-α	RA, PsA, AS, psoriasis; infliximab and adalimumab also Crohn and UC	TB, fungal, hepatitis B; heart failure; demyelination; lymphoma (boxed)
Tocilizumab, sarilumab	IL-6 receptor	RA; tocilizumab also giant cell arteritis	Infection, liver enzymes, lipids, cytopenias; no sarilumab with diverticulitis
Rituximab	CD20 B cells (depletion)	RA with methotrexate; GPA	First-dose rash (~30%), infusion reactions, hepatitis B reactivation
Abatacept	CD80/86–CD28 costimulation	RA, PsA, JIA	Infection; don't combine with TNF inhibitors; live vaccines
Anakinra, canakinumab, rilonacept	Interleukin-1	CAPS; anakinra approved but rarely used in RA	Injection-site reactions (up to 40%), URIs

Drug(s)	Target	Main uses	Watch for
Baricitinib, tofacitinib, upadacitinib	Janus kinase (JAK-STAT), oral	RA after TNF or methotrexate failure; PsA; UC	Boxed VTE/MACE/cancer; herpes zoster; ↑ LDL; caution with diverticulitis
Secukinumab, ixekizumab	IL-17A	Psoriasis, PsA, AS	Infection; can cause or worsen IBD; TB screen
Ustekinumab	IL-12/IL-23 (p40)	PsA, psoriasis, Crohn, UC	URIs; stop ≥ 15 weeks before live vaccines
Guselkumab, risankizumab	IL-23 (p19)	Plaque psoriasis, PsA	URIs; TB screen
Belimumab	BLyS (B-cell survival)	SLE, lupus nephritis	Infection; depression and suicide reported
Anifrolumab	Type 1 interferon receptor	Nonrenal SLE	URIs, bronchitis, infusion reactions

How to tell them apart. Sources: S28; p. 682; p. 683; p. 684; p. 685; p. 686; p. 689; p. 690; p. 692; p. 693

SLIDE DISCREPANCY

- **How methotrexate works at RA doses**
- Slides: Inhibits dihydrofolate reductase, reducing immune cell proliferation at low RA doses. S27; SG U2-M
- Textbook: Low-dose methotrexate mainly inhibits AICAR transformylase and thymidylate synthetase, raising extracellular AMP that suppresses inflammation; its dihydrofolate reductase effect is relatively minor at rheumatic doses. p. 688

WHY IT HAPPENS

- Why can't RA be treated like OA? RA is immune-mediated joint destruction; analgesics ease pain but leave the immune attack — and the damage — running. S23; S25 (notes)
- Why screen for TB and hepatitis first? These drugs suppress the immune response that keeps latent infections in check, so a hidden infection can reactivate. S30

MNEMONIC (MEMORY AID)

- Methotrexate rules = **W-F-P**: **W**eekly (never daily), **F**olate with it, **P**regnancy contraindicated. S27; SG U2-R7
- Methotrexate toxicity hits **L-B-L**: **L**iver, **B**one marrow, **L**ung. S27
- Generic-name endings: **-mab** = monoclonal antibody (adalimumab, infliximab, tocilizumab, rituximab); **-cept** = receptor-fusion protein (etanercept, abatacept). S28; p. 682; p. 690

HIGH-YIELD

- OA = degenerative → pain control; RA = autoimmune → suppress disease activity. S23

- OA ladder: acetaminophen → oral or topical NSAIDs → intra-articular steroids for a single-joint flare. S24
- DMARDs (conventional and biologic — methotrexate, adalimumab, etanercept, infliximab, tocilizumab, rituximab, abatacept) modify RA's disease course rather than only masking symptoms. S31; S25; SG U2-R6
- Methotrexate: DHFR inhibitor, **weekly**, with folate. S27; SG U2-R7
- Methotrexate toxicity: liver, marrow, lung; teratogenic. S27
- HCQ = milder RA; sulfasalazine = combinations; leflunomide = if MTX not tolerated. S26
- TNF: adalimumab, etanercept, infliximab · IL-6: tocilizumab · B cell: rituximab · T-cell costimulation: abatacept. S28; SG U2-R8
- Shared risk: immunosuppression → infection. S30; SG U2-R8
- Before biologics: TB + hepatitis B/C screening, update (live) vaccines. S30

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Pain by mechanism	Nociceptive · inflammatory · neuropathic S5
Chronic pain	Persists beyond expected healing, often > 3 months S5
4 pain mediators	Prostaglandins, bradykinin, histamine, cytokines (TNF-α, IL-1, IL-6) S6; SG U2-R1
Corticosteroid site	Phospholipase A_2 (upstream) → broad mediator suppression S8; SG U2-R2
NSAID site	COX → prostaglandins specifically S8; S14
Acetaminophen effects	Analgesic + antipyretic; minimal anti-inflammatory S10
Acetaminophen toxic metabolite	NAPQI (N-acetyl-p-benzoquinone imine) S11; S11 (notes)
Acetaminophen antidote	N-acetylcysteine — glutathione precursor S11; SG U2-R3
Hidden acetaminophen	Combination cold, flu, and sleep products — ask about ALL sources S12
COX-1	Housekeeping: gastric protection, renal blood flow S15; SG U2-R4
COX-2	Induced at inflammation; main inflammatory prostaglandin source S15; SG U2-R4
Ibuprofen	Common first-line for mild-moderate pain S16
Naproxen	Longer half-life, twice-daily S16
Diclofenac	Oral and topical S16
Indomethacin	More potent; more GI/CNS effects S16
Ketorolac	Short-term only; significant GI/renal risk S16
Celecoxib trade-off	↓ GI risk, ↑ CV risk, same renal risk S17; SG U2-R5
NSAID adverse effects	GI (ulcer, bleed) · renal (AKI, fluid) · CV (HTN, events) S19
NSAID cautions	GI bleed/ulcer, CKD, HF/CV disease, anticoagulant, late pregnancy S20
Triple whammy	NSAID + ACEi/ARB + diuretic → AKI S20 (notes)
NSAID + corticosteroid	↑ GI ulcer risk S20
OA vs. RA	Degenerative → pain/function vs. autoimmune → suppress disease S23

Prompt	Answer
OA drugs	Acetaminophen, oral/topical NSAIDs, intra-articular steroid for a single-joint flare S24
DMARD	Alters RA's course, not just symptoms S25; SG U2-R6
Conventional DMARDs	Methotrexate (anchor), hydroxychloroquine (milder), sulfasalazine (combos), leflunomide (MTX intolerant) S26
Methotrexate	DHFR inhibitor · weekly · folate · liver/marrow/lung · teratogenic S27; SG U2-R7
TNF inhibitors	Adalimumab, etanercept, infliximab S28; SG U2-R8
IL-6 inhibitor	Tocilizumab S28
B-cell therapy	Rituximab S28
T-cell costimulation	Abatacept S28; SG U2-M
DMARD shared risk	Immunosuppression → infection S30; SG U2-R8
Before biologics	TB + hepatitis B/C screen; update vaccines (live) first S30
Acetaminophen max (adults)	Generally ≤ 4 g/day; 15 g may be fatal p. 694
Aspirin vs. other NSAIDs	Aspirin: irreversible platelet COX (8–10 days); others reversible p. 675; p. 677
Ketorolac limits	Start IV/IM/intranasal; ≤ 120 mg/day; ≤ 5 days; halve if > 65, < 50 kg, renal impairment p. 694
Celecoxib selectivity	10–20× more selective for COX-2 p. 677
Methotrexate dose	7.5 mg weekly → 15–25 mg weekly p. 688
Hydroxychloroquine monitoring	Baseline eye exam; annual after 5 years p. 684
TNF inhibitor boxed warning	Malignancy, especially lymphoma p. 692
JAK inhibitor boxed warning	VTE (DVT/PE), MACE, cancer p. 686
JAK inhibitors	Baricitinib, tofacitinib, upadacitinib — oral; zoster, ↑ LDL p. 686
RA prednisone ceiling	≤ 7.5 mg daily long term, taper p. 693

Self-test

First, the lecture's own 0 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Original questions

Q1 · Mechanism · Musculoskeletal Pain & the Inflammatory Cascade

A drug blocks phospholipase A_2, so arachidonic acid is not released from membrane phospholipids. Which class is this, and how does its effect compare with an NSAID's?

- **A** A COX-2 inhibitor, sparing COX-1 housekeeping in the stomach
- **B** An NSAID, blocking the same COX step that celecoxib blocks
- **C** Acetaminophen, acting centrally on the hypothalamus
- **D** A biologic DMARD, neutralizing TNF-α outside the cell
- **E** A corticosteroid, suppressing a broader range of mediators than an NSAID

ANSWER: E

Corticosteroids block phospholipase A_2 upstream of arachidonic acid release, suppressing a broad range of mediators; NSAIDs act only at COX (S8; SG U2-R2). S8; SG U2-R2

- **A.** COX-2 inhibitors act at the COX step, not phospholipase A_2 (S8).
- **B.** NSAIDs block COX, downstream of arachidonic acid release (S8).
- **C.** Acetaminophen acts centrally; it does not block phospholipase A_2 (S10).
- **D.** Cytokine inhibitors act on a separate, parallel pathway (S7).

Q2 · Mechanism · Musculoskeletal Pain & the Inflammatory Cascade

Which set lists the inflammatory mediators the lecture names as sensitizing nociceptors in musculoskeletal pain?

- **A** Thromboxane, glutathione, NAPQI, and bradykinin
- **B** Prostaglandins, bradykinin, histamine, and cytokines
- **C** Prostaglandins, dopamine, serotonin, and insulin
- **D** Histamine, folate, uric acid, and cytokines
- **E** Cytokines, calcium, vitamin D, and histamine

ANSWER: B

Tissue injury releases prostaglandins, bradykinin, histamine, and cytokines (TNF-α, IL-1, IL-6), which sensitize nociceptors (S6; SG U2-R1). S6; SG U2-R1

- **A.** Glutathione and NAPQI belong to acetaminophen metabolism (S11).
- **C.** Dopamine, serotonin, and insulin are not on the lecture's mediator list (S6).
- **D.** Folate and uric acid are not pain mediators in this lecture (S6).
- **E.** Calcium and vitamin D belong to bone pharmacology (S2).

Q3 · Mechanism · Acetaminophen

In acetaminophen overdose, why is N-acetylcysteine effective?

- **A** It speeds renal excretion of unchanged acetaminophen
- **B** It blocks COX in the liver and lowers prostaglandin levels
- **C** It is a glutathione precursor that restores NAPQI detoxification
- **D** It inhibits the enzyme that forms NAPQI from the drug
- **E** It binds acetaminophen in the gut and prevents absorption

ANSWER: C

Overdose depletes glutathione so NAPQI accumulates; N-acetylcysteine replenishes glutathione to detoxify NAPQI (S11; SG U2-R3). S11; SG U2-R3

- **A.** The antidote works by refilling glutathione, not by renal excretion (S11).
- **B.** COX blockade is the NSAID mechanism, not the antidote's (S14).
- **D.** The lecture's mechanism is glutathione replenishment (SG U2-R3).
- **E.** The lecture explains the antidote through glutathione, not binding (S11).

Q4 · Clinical vignette · Acetaminophen

A 40-year-old with an acutely swollen, inflamed knee asks for the drug that best treats the inflammation itself. Why is acetaminophen not the best choice?

- **A** Its anti-inflammatory effect at the periphery is minimal
- **B** It only works when given intravenously in hospital
- **C** It causes more GI bleeding than the NSAIDs do
- **D** It is contraindicated in anyone with joint swelling
- **E** It has no analgesic effect in musculoskeletal pain

ANSWER: A

Acetaminophen relieves pain and fever centrally, but its peripheral anti-inflammatory effect is minimal — it is not a reliable anti-inflammatory agent (S10). S10

- **B.** No route restriction is taught; it is used orally first-line for mild OA pain (S24).
- **C.** GI bleeding is the NSAID risk (S19).
- **D.** No such contraindication is taught; the issue is weak anti-inflammatory action (S10).
- **E.** It does relieve mild-to-moderate pain (S10).

Q5 · Mechanism · NSAIDs & COX-2 Selective Inhibition

Which statement best explains why nonselective NSAIDs cause GI and renal adverse effects?

- **A** They block phospholipase A_2 and stop all mediator release
- **B** They deplete glutathione, which injures the gut and kidney
- **C** They inhibit COX-2, the only source of protective prostaglandins
- **D** They block TNF-α, which normally protects the stomach lining
- **E** They inhibit COX-1, which protects gastric mucosa and renal blood flow

ANSWER: E

COX-1 is the housekeeping enzyme that maintains gastric mucosal protection and normal renal blood flow; inhibiting it contributes to GI and renal adverse effects (S15; SG U2-R4). S15; SG U2-R4

- **A.** That is the corticosteroid site (S8).
- **B.** Glutathione depletion is acetaminophen overdose (S11).
- **C.** COX-2 is the main source of inflammatory prostaglandins (S15).
- **D.** TNF-α blockade is a biologic DMARD mechanism (S28).

Q6 · Clinical vignette · NSAIDs & COX-2 Selective Inhibition

A patient with a history of peptic ulcer bleeding and no cardiac disease needs an anti-inflammatory for osteoarthritis. The team considers celecoxib. What trade-off must be weighed?

- **A** Lower cardiovascular risk, but higher liver toxicity
- **B** Lower GI risk, but higher cardiovascular risk and similar renal risk
- **C** Higher GI risk, but lower cardiovascular and renal risk
- **D** Lower GI risk, plus kidney protection from COX-2 selectivity
- **E** No trade-off: celecoxib is safer in every organ system

ANSWER: B

Celecoxib lowers ulcer/bleeding risk but is associated with increased cardiovascular event risk, and its renal risk is similar to nonselective NSAIDs (S17; SG U2-R5). S17; S18; SG U2-R5

- **A.** Higher cardiovascular risk is the taught trade-off (S17).
- **C.** Reversed — celecoxib has lower GI and higher CV risk (S17-18).
- **D.** COX-2 selectivity does not protect the kidney (S17).
- **E.** The lecture stresses it is not a free lunch (S17 notes).

Q7 · Clinical vignette · NSAIDs & COX-2 Selective Inhibition

A patient on lisinopril (an ACE inhibitor) and hydrochlorothiazide starts daily naproxen for knee pain. Two weeks later, creatinine has risen sharply. What pattern explains this?

- **A** The “triple whammy”: NSAID + ACE inhibitor + diuretic causing AKI
- **B** A GI ulcer from NSAID plus diuretic, causing dehydration
- **C** Expected COX-2 renal protection failing under stress
- **D** Hepatotoxicity from NAPQI formed by naproxen metabolism
- **E** Bleeding from the NSAID plus the ACE inhibitor together

ANSWER: A

NSAIDs with ACE inhibitors/ARBs raise AKI risk, and adding a diuretic is the high-yield “triple whammy” AKI pattern (S20; S20 notes). S20; S20 (notes)

- **B.** NSAID + corticosteroid raises ulcer risk; this is a kidney injury pattern (S20).
- **C.** COX-2 selectivity does not protect the kidney, and naproxen is nonselective (S16-17).
- **D.** NAPQI comes from acetaminophen (S11).
- **E.** NSAID + anticoagulant raises bleeding; this is a renal pattern (S20).

Q8 · Mechanism · NSAIDs & COX-2 Selective Inhibition

Which nonselective NSAID does the lecture restrict to short-term use because of significant GI and renal risk?

- **A** Diclofenac
- **B** Ibuprofen
- **C** Celecoxib
- **D** Naproxen
- **E** Ketorolac

ANSWER: E

Ketorolac is for short-term use only because of significant GI/renal risk (S16). S16

- **A.** Diclofenac is noted for oral and topical formulations (S16).
- **B.** Ibuprofen is a common first-line choice for mild-moderate pain (S16).
- **C.** Celecoxib is COX-2 selective; its trade-off is cardiovascular risk (S17).
- **D.** Naproxen is noted for its longer half-life and twice-daily dosing (S16).

Q9 · Clinical vignette · NSAIDs & COX-2 Selective Inhibition

A patient takes low-dose aspirin for cardioprotection and starts regular ibuprofen for back pain. What is the textbook concern?

- **A** Ibuprofen blocks aspirin’s irreversible platelet inhibition
- **B** Ibuprofen converts aspirin to the toxic metabolite NAPQI
- **C** Both drugs together protect the kidney from injury
- **D** Ibuprofen speeds aspirin’s renal excretion fivefold
- **E** Aspirin makes ibuprofen COX-2 selective, raising CV risk

ANSWER: A

Concomitant ibuprofen antagonizes aspirin’s irreversible platelet inhibition, limiting its cardioprotective effect (p. 680). p. 680

- **B.** NAPQI comes from acetaminophen (S11).
- **C.** NSAIDs reduce renal perfusion; they don’t protect the kidney (S19).
- **D.** Not the textbook’s interaction (p. 680).
- **E.** Aspirin does not change ibuprofen’s selectivity (p. 680).

Q10 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A 66-year-old has knee pain worse with activity, no systemic symptoms, and mild disease. Which first-line drug does the lecture call reasonable?

- **A** Methotrexate to slow joint destruction
- **B** Acetaminophen for mild osteoarthritis pain
- **C** Adalimumab to block TNF-α signaling
- **D** Ketorolac taken daily as maintenance
- **E** Hydroxychloroquine for mild disease

ANSWER: B

Acetaminophen is a reasonable first-line for mild OA pain (S24). S23; S24

- **A.** Methotrexate is the anchor DMARD for RA, an autoimmune disease (S23; S26).
- **C.** Biologic DMARDs treat RA, not degenerative OA (S23; S28).
- **D.** Ketorolac is short-term only (S16).
- **E.** Hydroxychloroquine is a DMARD for milder RA (S26).

Q11 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

What is the key difference in treatment goals between osteoarthritis and rheumatoid arthritis?

- **A** OA aims at pain and function; RA aims to suppress disease activity
- **B** OA aims to suppress immunity; RA aims at pain relief only
- **C** RA aims at weight loss; OA aims at infection control
- **D** Both aim only at pain relief with NSAIDs
- **E** OA aims at cartilage regrowth with DMARDs

ANSWER: A

OA treatment targets pain control and function; RA treatment must suppress disease activity, not just pain (S23). S23; S23 (notes)

- **B.** Reversed (S23).
- **C.** Neither is the taught goal (S23-24).
- **D.** RA needs disease-modifying therapy (S23 notes).
- **E.** OA is managed mainly with analgesics/NSAIDs (S23 notes).

Q12 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A patient newly started on methotrexate for RA asks whether to take a tablet every morning "to keep levels steady." What is the correct counseling?

- **A** Take it daily with food to reduce stomach upset
- **B** Take it daily but skip the folate supplement
- **C** Take it only on days when joints are painful
- **D** Take it once weekly, because daily dosing can be fatal
- **E** Take it twice daily, like naproxen, for steady levels

ANSWER: D

Methotrexate is dosed weekly, not daily — a critical patient-safety point, since daily dosing can be fatal (S27; SG U2-R7). S27; SG U2-R7

- **A.** Daily dosing is the dangerous error (S27).
- **B.** Folate is co-prescribed to reduce toxicity, and dosing is weekly (S27).
- **C.** It is a scheduled weekly DMARD, not an as-needed analgesic (S25; S27).
- **E.** Twice-daily dosing describes naproxen (S16).

Q13 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

What is methotrexate's mechanism as taught in the lecture, and what is co-prescribed to reduce its toxicity?

- **A** TNF-α blockade; tuberculosis screening
- **B** Phospholipase A_2 blockade; calcium supplementation
- **C** Dihydrofolate reductase inhibition; folate supplementation
- **D** COX-2 inhibition; a proton pump inhibitor
- **E** Glutathione depletion; N-acetylcysteine

ANSWER: C

Methotrexate inhibits dihydrofolate reductase, reducing immune cell proliferation at low RA doses; folate is routinely co-prescribed (S27; SG U2-R7; SG U2-M). S27; SG U2-M; SG U2-R7

- **A.** TNF blockade is the biologic DMARD mechanism (S28).
- **B.** Phospholipase A_2 is the corticosteroid site (S8).
- **D.** COX-2 inhibition is celecoxib (S17).
- **E.** That is acetaminophen toxicity and its antidote (S11).

Q14 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A 29-year-old with active RA who is planning a pregnancy needs a DMARD. Which drug from the lecture is contraindicated?

- **A** Celecoxib, because it is a disease-modifying drug
- **B** Acetaminophen, because it is a DMARD
- **C** Sulfasalazine, because it is only used alone
- **D** Hydroxychloroquine, because of its poor safety profile
- **E** Methotrexate, because it is teratogenic

ANSWER: E

Methotrexate is contraindicated in pregnancy because it is teratogenic (S27). S27

- **A.** Celecoxib is an NSAID, not a DMARD (S17; S25).
- **B.** Acetaminophen is an analgesic, not a DMARD (S10; S25).
- **C.** Sulfasalazine is often used in combination regimens (S26).
- **D.** Hydroxychloroquine has a relatively favorable safety profile (S26).

Q15 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

Which single feature distinguishes a DMARD from an NSAID in rheumatoid arthritis?

- **A** It gives faster, stronger analgesia than any NSAID
- **B** It is always a biologic given by injection
- **C** It changes the disease course, not just the symptoms
- **D** It blocks COX more selectively than an NSAID
- **E** It carries no risk of infection or toxicity

ANSWER: C

A DMARD is intended to alter the course of RA — slowing or halting joint destruction — not just mask symptoms (S25; SG U2-R6). S25; SG U2-R6

- **A.** DMARDs are not defined by analgesic speed (S25).
- **B.** Conventional DMARDs such as methotrexate are not biologics (S26).
- **D.** COX blockade is the NSAID mechanism (S14).
- **E.** Immunosuppression and infection risk are shared across DMARDs (S30).

Q16 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

Before starting adalimumab for rheumatoid arthritis, which baseline step does the lecture specifically call for?

- **A** Adding an ACE inhibitor to protect the kidney
- **B** Avoiding all vaccines until therapy has begun
- **C** Checking for all hidden acetaminophen sources
- **D** Screening for tuberculosis and hepatitis B/C
- **E** Starting folate to prevent bone marrow toxicity

ANSWER: D

Biologics cause immunosuppression; baseline tuberculosis and hepatitis B/C screening comes before starting them (S30). S30

- **A.** Not a DMARD safety step; NSAID + ACEi raises AKI risk (S20).
- **B.** Vaccines, especially live vaccines, are updated before therapy (S30).
- **C.** That is acetaminophen safety counseling (S12).
- **E.** Folate is co-prescribed with methotrexate (S27).

Q17 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

Which biologic DMARD works by blocking the second (costimulatory) signal needed for T-cell activation?

- **A** Rituximab
- **B** Infliximab
- **C** Tocilizumab
- **D** Etanercept
- **E** Abatacept

ANSWER: E

Abatacept is the T-cell costimulation modulator (S28; S29; SG U2-M). S28; S29; SG U2-M

- **A.** Rituximab depletes or modulates B lymphocytes (S28).
- **B.** Infliximab blocks TNF-α signaling (S28).
- **C.** Tocilizumab blocks IL-6 receptor signaling (S28).
- **D.** Etanercept blocks TNF-α signaling (S28).

Q18 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A patient on a biologic DMARD calls with a fever of 38.6 °C and a productive cough. What lecture principle drives the advice to be seen promptly?

- **A** NSAID-like renal toxicity makes fever dangerous
- **B** Biologics raise cardiovascular risk like celecoxib
- **C** Immunosuppression raises infection risk across all DMARDs
- **D** Biologics cause hidden acetaminophen overdose
- **E** Fever means the biologic is working as intended

ANSWER: C

The major safety theme across all DMARDs is immunosuppression → increased infection risk; patients should report fevers or infection symptoms promptly (S30; SG U2-R8). S30; SG U2-R8

- **A.** Renal toxicity is an NSAID theme, not the DMARD safety theme (S19; S30).
- **B.** That trade-off belongs to celecoxib (S17).
- **D.** Hidden acetaminophen comes from combination cold products (S12).
- **E.** Fever signals possible infection in an immunosuppressed patient (S30).

Q19 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

Match the drug to its target: which pairing is correct?

- **A** Abatacept — B-lymphocyte depletion
- **B** Tocilizumab — IL-6 receptor signaling
- **C** Rituximab — TNF-α signaling
- **D** Adalimumab — T-cell costimulation
- **E** Etanercept — IL-6 receptor signaling

ANSWER: B

Tocilizumab is the lecture's IL-6 inhibitor (S28; SG U2-R8). S28; SG U2-R8

- **A.** Abatacept modulates T-cell costimulation (S28).
- **C.** Rituximab depletes B cells; TNF is adalimumab, etanercept, infliximab (S28).
- **D.** Adalimumab blocks TNF-α (S28).
- **E.** Etanercept blocks TNF-α (S28).

Q20 · Clinical vignette · Osteoarthritis, Rheumatoid Arthritis & DMARDs

A patient with RA and high cardiovascular risk is being considered for tofacitinib. Which boxed warning applies to all JAK inhibitors?

- **A** Fatal hepatotoxicity from NAPQI accumulating in the liver
- **B** Lymphoma only, exactly the same as the TNF inhibitors
- **C** Retinopathy that requires annual eye examinations
- **D** Venous thromboembolism, major cardiovascular events, and cancer
- **E** Teratogenicity that requires strict weekly-only dosing

ANSWER: D

All JAK inhibitors carry an FDA boxed warning for VTE (DVT, PE), MACE, and cancer (p. 686). p. 686

- **A.** NAPQI is acetaminophen toxicity (S11).
- **B.** The TNF inhibitor warning is malignancy; JAK inhibitors add VTE and MACE (p. 686; p. 692).
- **C.** Retinal screening applies to hydroxychloroquine (p. 684).
- **E.** Weekly dosing is a methotrexate rule (S27).

Q21 · Mechanism · Osteoarthritis, Rheumatoid Arthritis & DMARDs

Which conventional DMARD requires a baseline eye exam and annual screening after 5 years because of retinopathy?

- **A** Leflunomide
- **B** Sulfasalazine
- **C** Hydroxychloroquine
- **D** Methotrexate
- **E** Mycophenolate mofetil

ANSWER: C

Antimalarials bind melanin-rich tissue such as the retina; a baseline exam and annual screening after 5 years are recommended (p. 684). p. 684

- **A.** Leflunomide's issues are diarrhea and liver tests (p. 688).
- **B.** Sulfasalazine's issues are GI, rash, and cytopenias (p. 690).
- **D.** Methotrexate monitoring targets liver, marrow, and lung (S27).
- **E.** Mycophenolate's issues are GI, liver, cytopenias, and infection (p. 689).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO2.1** Identify major pharmacologic classes used in the management of musculoskeletal conditions and describe their mechanisms of action.
- [ ] **IO2.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat musculoskeletal conditions.
- [ ] **IO2.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO2.4** Select appropriate pharmacologic therapy for common musculoskeletal conditions based on clinical presentation and patient-specific factors.
- [ ] **IO2.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in musculoskeletal conditions.
- [ ] **IO2.6** Explain important medication-use and safety information to patients receiving pharmacotherapy for musculoskeletal.

- [ ] **SO1** Explain the pharmacologic basis of musculoskeletal pain and inflammation.
- [ ] **SO2** Compare acetaminophen and NSAIDs, including COX-1 vs. COX-2 inhibition.
- [ ] **SO3** Identify major NSAID adverse effects and select therapy based on patient risk factors.
- [ ] **SO4** Explain the mechanisms of conventional vs. biologic DMARDs, including representative generic agents in each class.
- [ ] **SO5** Identify key contraindications, adverse effects, and drug interactions for analgesics and DMARDs.
- [ ] **SO6** Preview: gout, skeletal muscle relaxants, and bone pharmacology continue in Session 2 (Thursday, October 29).